

Insurance Charges Prediction

Objective:

Build and train a regression model to predict insurance charges for customers using **Azure Machine Learning**, and deploy the model for real-time predictions.

Steps performed:

Create Resource group

Home > Resource Manager | Resource groups >

Create a resource group

Automation Link

Basics

Subscription	Azure subscription 1
Resource group name	rg_azure
Region	East US

Tags

None

Previous Next **Create**

Azure ML→Create workspace

Home > Azure Machine Learning >

Azure Machine Learning

Create a machine learning workspace

Validation passed

Basics Inbound Access Outbound Access Encryption Identity Tags **Review + create**

Subscription	Azure subscription 1
Resource group	rg_azure
Region	East US
Name	ws_azure
Storage account	(new) wsazure5284963786
Key vault	(new) wsazure9426880684
Application insights	(new) wsazure9876887849
Container registry	None
Connectivity method	Enable public access from all networks
Network isolation	Public
Encryption type	Microsoft-managed keys

Identity: none System: assigned

Create < Previous Next > Download a template for automation

Home > Microsoft.MachineLearningServices | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.MachineLearningServices
Subscription : Azure subscription 1
Resource group : rg_azure

Start time : 2/2/2026, 7:38:31 AM
Correlation ID : 532197fd-f752-4158-8532-d77b039d44e8

> Deployment details

> Next steps

Go to resource

Deployment succeeded

Deployment 'Microsoft.MachineLearningServices' to resource group 'rg_azure' was successful.

Go to resource Go to resource group

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

Find an Azure expert >

Add or remove favorites by pressing Ctrl+Shift+F

Launch ML Studio

Microsoft Azure Upgrade

Search resources, services, and docs (G+)

Copilot

santhoshc1902@gmail.com

Home > Microsoft.MachineLearningServices | Overview

ws_azure

Azure Machine Learning workspace

Suggest a workload for this ML workspace

Search

Download config.json Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Events

Settings

Monitoring

Automation

Support + troubleshooting

Essentials

Resource group : rg_azure

Location : East US

Subscription : Azure subscription 1

Storage : wsazure5284963786

Provisioning State : Succeeded

Studio web URL : https://ml.azure.com?tid=c064b21c-06dc-409b-a09e-c1e664eba...

Container Registry : ...

Key Vault : wsazure9426880684

Application Insights : wsazure9876887849

MLflow tracking URI : azureml://eastus.api.azureml.ms/mlflow/v1.0/subscriptions/04a2...

JSON View

Work with your models in Azure Machine Learning Studio

The Azure Machine Learning Studio is a web app where you can build, train, test, and deploy ML models. Launch it now to start exploring, or learn more about the Azure Machine Learning studio

Launch studio

Add or remove favorites by pressing Ctrl+Shift+F

Create Compute

Microsoft Foundry | Azure Machine Learning

Create compute instance

1 Required settings

2 Scheduling optional

3 Security optional

4 Applications optional

5 Tags optional

6 Review

Virtual machine type

CPU GPU

Virtual machine size

Select from recommended options Select from all options

Name ↑	Category	Workload types	Available quota	Cost
Standard_DS11_v2 2 cores, 14GB RAM, 28GB storage	Memory optimized	Development on Notebooks (or other IDE) and light weight testing	6 cores	\$0.18/hr
Standard_DS3_v2 4 cores, 14GB RAM, 28GB storage	General purpose	Classical ML model training on small datasets	6 cores	\$0.29/hr
Standard_E4ds_v4 4 cores, 32GB RAM, 150GB storage	Memory optimized	Data manipulation and training on medium-sized datasets (1-10GB)	4 cores	\$0.29/hr

Review + Create Back Next Cancel

Microsoft Foundry | Azure Machine Learning

Default Directory > ws_azure > Compute

Compute

The "Kubernetes clusters" tab is now where you can access previous versions of "inference clusters" (also known as "AKS clusters") and "attached Kubernetes" compute types along with any previously created compute targets using those types. [Learn more about Kubernetes clusters.](#)

Compute instances Compute clusters Kubernetes clusters Attached computes Serverless instances

Choose from a selection of CPU or GPU instances preconfigured with popular tools such as VS Code, JupyterLab, Jupyter, and RStudio, ML packages, deep learning frameworks, and GPU drivers. [Learn more about compute instances](#)

[+ New](#) [Refresh](#) [Start](#) [Stop](#) [Restart](#) [Schedule and idle shutdown](#) [Delete](#) [Reset view](#) [View quota](#)

Search

Name	State	Idle shutdown	Applications	Size
compute-ml1	Running	1 hour	JupyterLab Jupyter VS Code (Web) ...	Standard

Data Load

Microsoft Foundry | Azure Machine Learning

Create data asset

1 Data type 2 Data source

Set the name and type for your data asset

Name *
insurance

Description
Insurance Dataset

Type *
Tabular

Use cases for data types

When should I use File type?

The File type is recommended in most scenarios when you are working with a single data file of any type (including tabular data). This type allows you to specify a file location by URI in a storage location on your local computer, an attached Datasore, blob/ADLS storage, or a publicly available http(s) location. There are many types of supported URIs. In the Azure Machine Learning CLI v2 or Python SDK v2, this data type is called uri_file. [Learn more about the uri_file type](#)

When should I use Folder type?

The Folder type has all the same capabilities and use

[Back](#) [Next](#) [Cancel](#)

Microsoft Foundry | Azure Machine Learning

Default Directory > ws_azure > Data > insurance

insurance

Version: 1 (latest) ☆

Details Consume Explore Models Jobs

[New version](#) [Refresh](#) [Generate profile](#) [Archive](#)

Attributes

Type
Table (mtable)

Dataset type (from Azure ML v1 APIs)
Tabular

Created by
Santhosh Chinnasamy

Profile
[View profile](#)
Job: --

Files in dataset
1

Total size of files in dataset
116.1 KiB

Tags

No data

Description
Insurance Dataset

Data sources

Datasore
[workspaceblobstore](#)

Relative path
UI/2026-02-02_021743_UTC/insurance_data.csv

Actions

Model Training

Automated ML

Microsoft Foundry | Azure Machine Learning

Default Directory > ws_azure > Training job

Submit an Automated ML job

1 Training method
2 Basic settings
3 Task type & data
4 Task settings
5 Compute
6 Review

Basic settings

Let's start with some basic information about your training job.

Job name *

Experiment name *
☐ Select existing ☒ Create new

New experiment name *

Description

Back Next Cancel

Microsoft Foundry | Azure Machine Learning

Default Directory > ws_azure > Training job

Submit an Automated ML job

1 Training method
2 Basic settings
3 Task type & data
4 Task settings
5 Compute
6 Review

Task type & data

Select task type *

Select data

Make sure your data is preprocessed into a supported format.

[+ Create](#) [Refresh](#) ☒ Show supported data assets only [Reset view](#)

Search

Name	Type	Created on	Modified on
insurance	Table	Feb 2, 2026 7:49 AM	Feb 2, 2026 7:49 AM

Page 1 of 1 25/Page

Back Next Cancel

Microsoft Foundry | Azure Machine Learning

Default Directory > ws_azure > Training job

Submit an Automated ML job

1 Training method
2 Basic settings
3 Task type & data
4 Task settings
5 Compute
6 Review

Task settings

Task type
Regression

Data
insurance (View data)

Target column *

[View additional configuration settings](#) [View featurization settings](#)

Limits

Max trials

Back Next Cancel

Microsoft Foundry | Azure Machine Learning

Default Directory > ws_azure > Training job

Submit an Automated ML job

Training method

Basic settings

Task type & data

Task settings

Compute

Review

Max concurrent trials ①

2

Max nodes ①

Enter max nodes

Metric score threshold ①

Enter metric score threshold

Experiment timeout (minutes) ①

30

Iteration timeout (minutes) ①

Enter timeout in minutes

☐ Enable early termination ①

Validate and test

Back

Next

Cancel

Microsoft Foundry | Azure Machine Learning

Default Directory > ws_azure > Training job

Submit an Automated ML job

Training method

Basic settings

Task type & data

Task settings

Compute

Review

You can choose a validation type and select test data as an optional step.

Validation type ①

Train-validation split

Percentage validation of data * ①

20

Automated ML recommends that between 10 and 30 percent of data is held out for validation

Test data ①

Train-test split

Percentage test of data * ①

20

Automated ML recommends that between 10 and 30 percent of data is held out for test

Back

Next

Cancel

Microsoft Foundry | Azure Machine Learning

Default Directory > ws_azure > Training job

Submit an Automated ML job

Training method

Basic settings

Task type & data

Task settings

Compute

Review

Compute

Select and configure the compute resource for executing your training job.

Select compute type

Compute instance

Select Azure ML compute instance *

compute-ml1, Running, Standard_DS3_v2, 4 vCPUs (cores), 14 GB, 28 GB (storage), \$0.29...

+ New

Back

Next

Cancel

Microsoft Foundry | Azure Machine Learning

Default Directory > ws_azure > Training job

Submit an Automated ML job

Training method

Basic settings

Task type & data

Task settings

Compute

Review

Review

Review or make changes to your job before submission.

Basic settings

Name
auto-ml

Experiment name
experiment1

Timeout (hours)
--

Task type & data

Task type
Regression

Data
insurance

Task settings

Target column
charges

Max trials: --
Max concurrent runs: --
Max nodes: --
Metric score
Experiment iteration time
Validate type
Train-validation-test
Percentage

Back

Submit training job

Cancel

Microsoft Foundry | Azure Machine Learning

Default Directory > ws_azure > Jobs > experiment1 > auto-ml

auto-ml

Completed

Overview

Data guardrails

Models + child jobs

Outputs + logs

Child jobs

Refresh

Edit and submit (preview)

Register model

Cancel

Delete

Compare

Properties

Status
Completed

Created on
Feb 2, 2026 7:58 AM

Start time
Feb 2, 2026 7:59 AM

Duration
34m 20.59s

Compute duration
34m 20.59s

Compute target

Inputs

Input name: training_data
Data asset: insurance:1
Asset URI: azureml:insurance:1

Outputs

Output name: best_model
Model: azureml_auto-ml_79_output_mlflow_log_model_58387562:1
Asset URI: azureml:azureml_auto-ml_79_output_mlflow_log_model_5838...

Best model summary

Microsoft Foundry | Azure Machine Learning

Default Directory > ws_azure > Jobs > experiment1 > auto-ml

auto-ml

Completed

Overview

Data guardrails

Models + child jobs

Outputs + logs

Child jobs

Refresh

Deploy

Download

Explain model

View generated code

Reset view

Search

Filter

Columns

Algorithm name	Responsible AI	Normalized ro...	Sampling	Created on
StandardScalerWrapper, XGBoostRegressor		0.01116	100.00 %	Feb 2, 2026 8:02 AM
MaxAbsScaler, XGBoostRegressor		0.01131	100.00 %	Feb 2, 2026 8:02 AM
StandardScalerWrapper, XGBoostRegressor		0.01171	100.00 %	Feb 2, 2026 8:24 AM
MaxAbsScaler, LightGBM		0.01192	100.00 %	Feb 2, 2026 8:23 AM
MaxAbsScaler, ExtremeRandomTrees		0.01212	100.00 %	Feb 2, 2026 8:28 AM

<<

<

Page 1 of 4

>

>>

25/Page

Model Selection

XGBoost Regressor has been selected and registered

Microsoft Foundry | Azure Machine Learning

ws_azure > Jobs > experiment1 > auto-ml > frank_moon_c5wvgs3k

frank_moon_c5wvgs3k Completed

Overview **Model** Metrics Responsible AI (preview) Data transfer

Refresh Deploy Download Explain model

Model summary

Algorithm name
StandardScalerWrapper, XGBoostRegressor

Hyperparameters
[View hyperparameters](#)

Normalized root mean squared error
0.01116 [View all other metrics](#)

Sampling
100.00 %

Registered models
No registration yet

Deploy status

Run Metrics

Explained variance
0.99663

Mean absolute error
493.36

Mean absolute percentage error
4.8932

Median absolute error
294.55

Normalized mean absolute error
0.0078751

Normalized median absolute error
0.0047017

Normalized root mean squared error
0.011162

Normalized root mean squared log error
0.016306

R2 score
0.99662

Root mean squared error
699.30

Register model

Microsoft Foundry | Azure Machine Learning

ws_azure

Register model from a job output

Select job
Select output
Model settings
Review

Select output
Specify the corresponding output to register the model.

A named model output has been detected in the job outputs and auto-selected. You can select any other output and/or type if this is not what you intend to select.

Model type *
MLflow

Job output *
mlflow_log_model_1414330808 (azureml-auto-ml_19_output_mlflow_log_mo...)

Back Next Cancel

Microsoft Foundry | Azure Machine Learning

ws_azure

Register model from a job output

Select job
Select output
Model settings
Review

Review
Review or make changes to your selections.

Select job
Job
frank_moon_c5wvgs3k

Select output
Model type
MLflow
Named model output
mlflow_log_model_1414330808 (azureml-auto-ml_19_output_mlflow_log_model_1414330808:1)

Model settings
Name
XGBoost-regressor
Description
--
Version
--
Tags
No tags

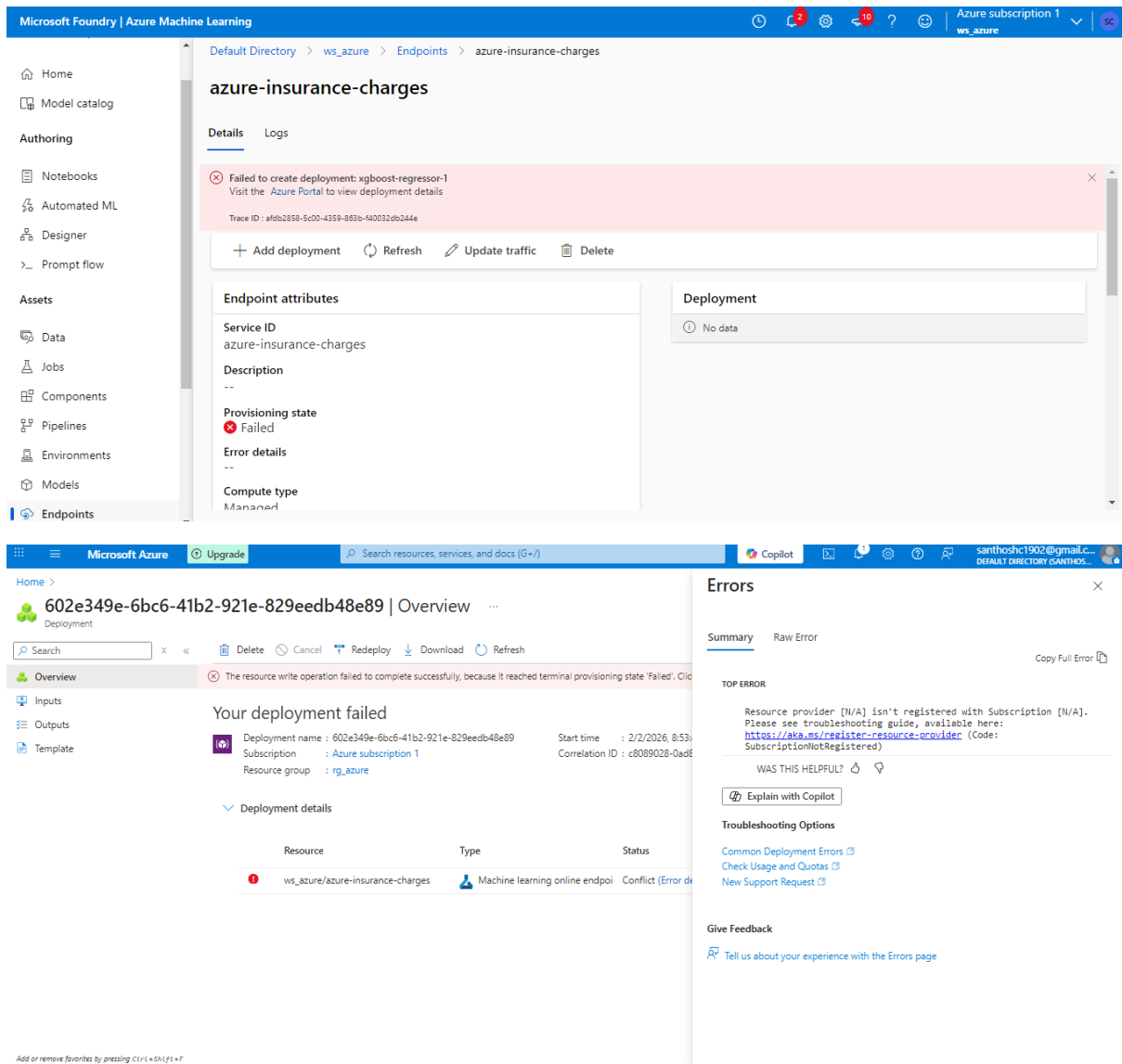
Back Register Cancel

Model Deployment

The screenshot shows the Microsoft Foundry | Azure Machine Learning interface. The left sidebar contains navigation options: Home, Model catalog, Authoring, Notebooks, Automated ML, Designer, Prompt flow, Assets, Data, Jobs, Components, Pipelines, Environments, Models (selected), and Endpoints. The main panel displays the details for the model 'XGBoost-regressor:1'. The 'Attributes' section lists: Name (XGBoost-regressor), Version (1), Created on (Feb 2, 2026 8:51 AM), Created by (Santhosh Chinnasamy), Type (MLFLOW), and Created by job (auto-ml_19). A dropdown menu is open, showing options: Real-time endpoint (Deploy the model using the real-time endpoint wizard), Batch endpoint (Deploy the model using the batch endpoint wizard), and Web service (Deploy to a web service (only for models based on frameworks)). The right sidebar shows 'Tags' (No tags), 'Properties' (No properties), and 'Description' (Click edit icon to add a description).

The screenshot shows the 'Deploy XGBoost-regressor:1' dialog box. The 'Project Name' is 'ws_azure'. The 'Instance count' is set to 1. The 'Virtual machine' is set to 'Standard_D2as_v4' (2 Cores, 8 GB (RAM), 16 GB (Disk), \$0.10/hr). The 'Endpoint' is set to 'New'. The 'Endpoint name' is 'azure-insurance-charges'. A note states: 'An endpoint URL will be generated after creating an endpoint. https://azure-insurance-charges.eastus.inference.ml.azure.com/score Learn how to consume'. The 'Deployment name' is 'xgboost-regressor-1'. The 'Inferencing data collection' is set to 'Disabled'. The 'Package Model' is set to 'Disabled'. The 'Deploy' button is highlighted.

The screenshot shows the 'Deploy XGBoost-regressor:1' dialog box. The 'Project Name' is 'ws_azure'. The 'Instance count' is set to 1. The 'Virtual machine' is set to 'Standard_D2as_v4' (2 Cores, 8 GB (RAM), 16 GB (Disk), \$0.10/hr). The 'Endpoint' is set to 'New'. The 'Endpoint name' is 'azure-insurance-charges'. A note states: 'An endpoint URL will be generated after creating an endpoint. https://azure-insurance-charges.eastus.inference.ml.azure.com/score Learn how to consume'. The 'Deployment name' is 'xgboost-regressor-1'. The 'Inferencing data collection' is set to 'Disabled'. The 'Package Model' is set to 'Disabled'. The 'Deploy' button is highlighted.



Cloud Deployment Workaround

Due to limitations in the Azure ML free trial, deployment of the model and generation of the endpoint/API key was not possible. As a workaround:

- The regression model was trained in Jupyter Notebook and saved as a .pkl file
- The saved model was then deployed on Hugging Face Spaces using Gradio for an interactive web interface

Live demo: [Insurance Charges Prediction](#)

Insurance Charges Predictor

Predict insurance charges based on patient and hospital data

Smoker

yes

Claim Amount

0

Past Consultations

0

Num of Steps

0

Hospital Expenditure

0

Number of Past Hospitalizations

0

Annual Salary

0

Clear

Submit

Predicted Charges

0

Share via Link

Examples

Smoker	Claim Amount	Past Consultations	Num of Steps	Hospital Expenditure	Number of Past Hospitalizations	Annual Salary
no	29087.54313	17	715428	4720205.992	0	55784970.05
no	39053.67437	7	699157	4329831.676	0	13700885.19